WebClient Computing

HTML Basics

Department of Computer Science

Kyungyong Lee

Project Rules

- How about copying other existing pages?
  - Encourage you to reference other pages
  - Read the others code, understand it, and apply it at your site
  - Whether you understand tags/elements in the page will be evaluated during the interview with the instructor
  - Honestly, junior level software engineers do not start programming from scratch
  - Add links of pages that you referenced
    - Referencing many pages are absolutely all right as long as you understand them!!
    - Copyright issue, Ethics
    - You will get in trouble if you omit references that you get hints
- Do not use any UI tools that automate the page creation. Write HTML/CSS code by yourself
  - Automatically generated codes are easily noticeable

<td> and <th> in <table> Tag

- <th> implies title of each column (example: 이름)
- <td> is an element (example: 이경용)

HTML5 Details

- <details> tag is useful when we want to hide details to be shown
- <summary> tag nested within <details> specifies a string to be shown

```html
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 Details</title>
</head>
<body>
  <details>
    <summary>KMU CS</summary>
    Kookmin University Computer Science
  </details>
</body>
</html>
```

▶ KMU CS

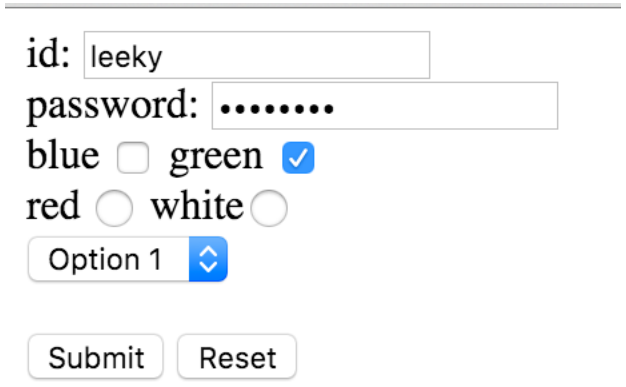
▼ KMU CS
Kookmin University Computer Science

HTML Getting Input

- A client can send information to a server
- Input types
  - text – use a plain text. Using value attribute, a default value can be set
  - password – to get password input (hidden value)
  - checkbox – check box that the selection can be toggled
  - radio – user can select only one
  - select – expresses a drop down menu

HTML Getting Input Example

```html
<!DOCTYPE html>
<html>
<head>
  <title>HTML input type</title>
</head>
<body>
  <form>
    id: <input type="text" value="leeky"><br>
    password: <input type="password" value="password"><br>
    blue <input type="checkbox"> green <input type="checkbox" checked><br>
    red <input type="radio" name="color"> white<input type="radio" name="color"><br>
    <select>
      <option>Option 1</option>
      <option>Option 2</option>
    </select><br><br>
    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
  </form>
</body>
</html>
```



HTML5 Input Types

- <tel> : to get a telephone number
- <email> : to get an email address
- <url> : to get an URL as an input
- <datetime> : to get date and time format input
- <number> : get a number type input
- <range> : a range of numbers is shown in a form of slide bar
- <color> : shows color as an option

HTML5 Input Types

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 input type</title>
</head>
<body>
  <form>
    email : <input type="email" name="myemail" required=""><br>
    url : <input type="url" name="myurl" required=""><br>
    phone : <input type="phone" name="myphone" maxlength="11"><br>
    date : <input type="date" name="mydatetime"><br>
    number : <input type="number" name="mynumber" min="10" max="20"><br>
    range : <input type="range" name="myrange" min="30" max="40"><br>
    color : <input type="color" name=mycolor"><br>
    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
  </form>
</body>
</html>
```

HTML Block and Inline Elements

- Block elements always start on a new line and use the full width
  - , <h1> ~ <h6>, <p>
  - can embed other html file and have own styles
- Inline elements does not start on a new line and does not occupy the full width
  - <a>, <img>,
  - is generally used as a container for a text

and Example

```
<!DOCTYPE html>
<html>
<body>

<div style="background-color:black; color:white; padding:20px;">
<h2>London</h2>
<p>London is the capital city of England. It is the most populous city in the United Kingdom,
with a metropolitan area of over 13 million inhabitants.</p>
</div>

<h1>My <span style="color:red">Important</span> Heading</h1>

</body>
</html>
```

London

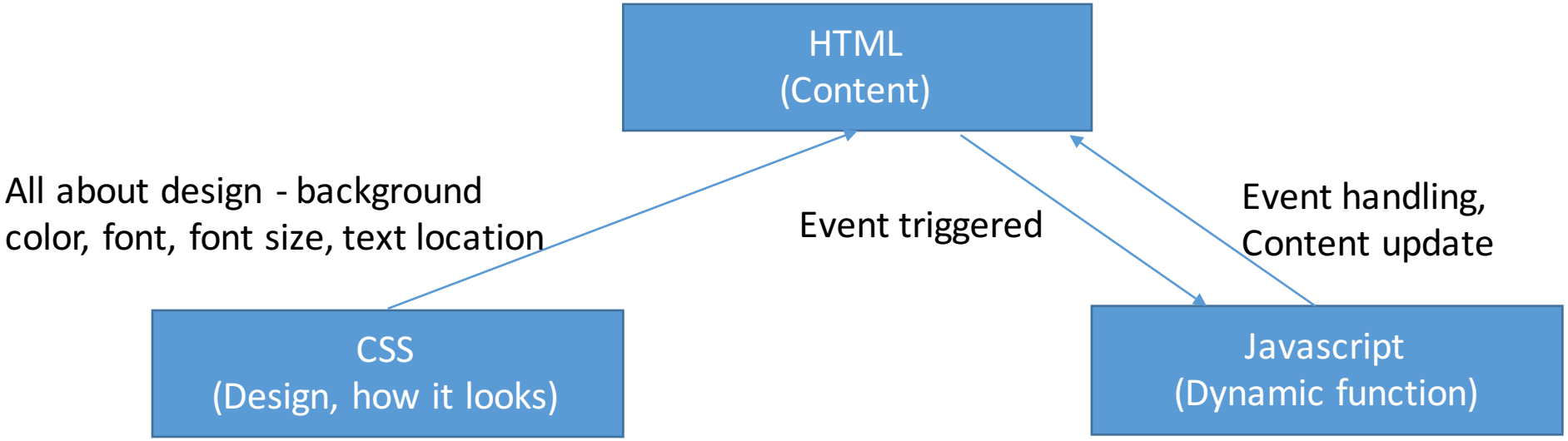
London is the capital city of England. It is the most populous city in the United Kingdom, with a metropolitan area of over 13 million inhabitants.

My Important Heading

HTML Tag References

- http://www.w3schools.com/tags/
- We covered only fraction of them. You are encouraged to use them in you project!!!

HTML, CSS, and Javascript



- HTML – Hyper Text Markup Language
  - House itself
- CSS - Painting the house and decoration
- Javascript – rich features, electricity on/off using a smartphone

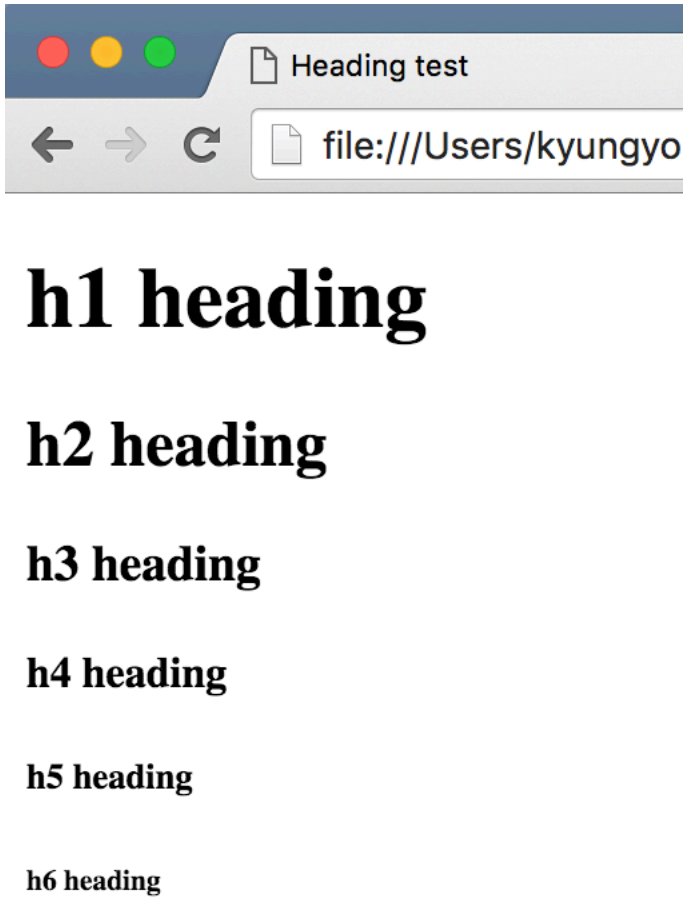
CSS – Cascading Style Sheets

- Decides how HTML file is presented in a web browser
- Remember <h1> ~ <h6> example
- <h1> ~ <h6> means hierarchical content organization
- It does not specify font size and color!
- Using CSS we can change the format

HTML Headings

- Organizing contents hierarchically
  - Recommended to use in order
  - h1, h2, h3, h4, h5, h6
  - Space is automatically added after heading
  - Search engine use headings to **index** the structure of the content

```html
<!DOCTYPE html>
<html>
<head>
  <title>Heading test</title>
</head>
<body>
  <h1>h1 heading</h1>
  <h2>h2 heading</h2>
  <h3>h3 heading</h3>
  <h4>h4 heading</h4>
  <h5>h5 heading</h5>
  <h6>h6 heading</h6>
</body>
</html>
```



Same HTML, Different Outlook

h1 heading

h2 heading

h3 heading

h4 heading

h5 heading

h6 heading

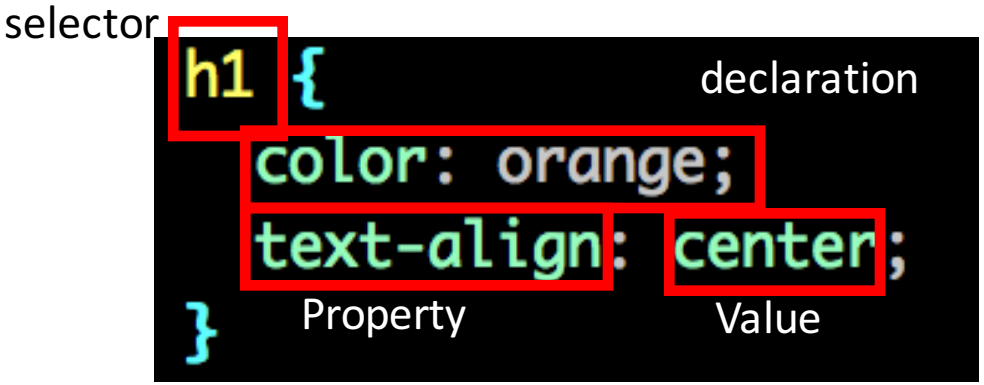
```html
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      color: orange;
      text-align: center;
    }
    h2 {
      color: blue;
      text-align: left;
    }
    h3 {
      color: green;
      text-align: left;
      font-size: xx-large;
    }
  </style>
  <title>Heading test</title>
</head>
<body>
  <h1>h1 heading</h1>
  <h2>h2 heading</h2>
  <h3>h3 heading</h3>
  <h4>h4 heading</h4>
  <h5>h5 heading</h5>
  <h6>h6 heading</h6>
</body>
</html>
```

Benefits of CSS

- Separation of content from its presentation – modular
- Reusable style by having an external CSS file
- Useful to show HTML in different design and layout
  - Desktop version and mobile device version
- Without CSS, embedding style information in a HTML file makes the management very hard
  - Adding font-size in the HTML

CSS Syntax

- CSS Syntax consists of
  - Selector – HTML elements where the style will be applied to
    - Tag, ID (starts from #), class (starts from .)
  - Declaration – specifies style to be applied to the selector. Multiple attributes separated by semi-colon;
    - Property: The name of the attribute (ex. font-size)
    - Value: The value of the attribute (ex. xx-large)
    - Property and Value is separated by colon



How to Insert Style Sheets

- External
  - An external style sheet file is created with an extension of .css and is referenced within head

```html
<!DOCTYPE html>
<html>
<head>
 <link rel="stylesheet" type="text/css" href="styles.css">
  <title>My first web page</title>
</head>
```

- Internal
  - Inserted within header <style> tag

```html
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      color: orange;
      text-align: center;
    }
  </style>
  <title>Heading test</title>
</head>
```

How to Insert Style Sheets

- Inline styles
  - A style can be added for a single element within a start tag

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <h1 style="color: orange;text-align: center;">h1 heading</h1>
</body>
</html>
```

Meaning of Cascading

- Cascade means “폭포”
- The style rule is applied in the order of
  - Inline styles – the highest order
  - External and internal style sheet in the <head> section
  - Browser default

References

- http://www.w3schools.com/html/ - HTML tutorial
- http://www.htmldog.com/guides/html/
- http://fromyou.tistory.com/581
- http://www.w3schools.com/css/default.asp - CSS Tutorial